TOPIC 06 · DIMENSIONAL ANALYSIS

Dimensional Analysis

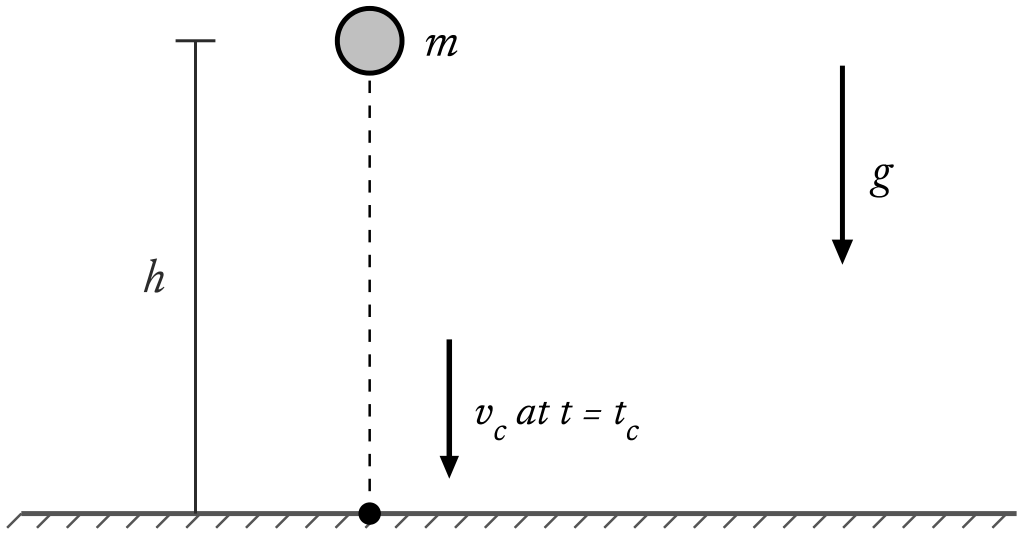
Dimensions and independent dimensions, the Π theorem, Taylor's explosion, the parameters read from the equations, the nondimensional Navier-Stokes equations, the limits Re ≫ 1 and Re ≪ 1, physical similarity, and a practical formulary.

> **Central idea.** A physical law cannot depend on the units chosen to measure its quantities. If an unknown a_0 depends on n governing parameters, of which k have independent dimensions, the Π theorem reduces the law to a relation between $n-k+1$ dimensionless groups, $\Pi_0 = F(\Pi_1, \ldots, \Pi_{n-k})$. In fluid mechanics the list of parameters is read from the conservation equations and their boundary conditions; the limits $\mathrm{Re} \gg 1$ and $\mathrm{Re} \ll 1$ remove one more parameter; and equal dimensionless groups in a model and in the real system (physical similarity) let a laboratory test predict the full-scale result.

1. Dimensions of a Physical Magnitude

The falling body

The chapter opens with a mass m released from rest at a height h under gravity g (Notes.pdf, p. 57, Fig. 6.1). The unknowns are the collision time t_c and the collision velocity v_c.



unknowns t_c and v_c, functions of (m, h, g): Eq. (6.1)

Figure 1. The fall of a point mass, Fig. 6.1 of the notes: a mass m released from rest at a height h falls under gravity g; the unknowns are the collision time t_c and the collision velocity v_c.

The exact solution is quick, and it will be used to check the dimensional result. Newton's second law $m\,d^2z/dt^2 = -mg$ with $z = h$ and $dz/dt = 0$ at $t = 0$ is integrated once, from 0 to t, using Barrow's rule:

$$\frac{dz}{dt}(t) - \frac{dz}{dt}(0) = \int_0^t (-g)\,dt' = \big[-gt'\big]_0^t = -gt \quad\Longrightarrow\quad \frac{dz}{dt} = -gt$$

and once more:

$$z(t) - z(0) = \int_0^t (-gt')\,dt' = \left[-\frac{gt'^2}{2}\right]_0^t = -\frac{gt^2}{2} \quad\Longrightarrow\quad z = h - \frac{gt^2}{2}$$

Setting $z(t_c) = 0$ gives $t_c = \sqrt{2h/g}$, and then $v_c = gt_c = \sqrt{2gh}$ (Notes.pdf, p. 57).

What we would guess before solving

Before any integration, one guesses which quantities can influence the answer: the mass, the height and gravity (Notes.pdf, Eq. 6.1):

$$t_c = t_c(m, h, g), \qquad v_c = v_c(m, h, g) \tag{6.1}$$

Units and dimensions

The numbers t_c, v_c, h, m, g are the result of comparing each quantity with a unit. If the unit of length is made L times smaller, the unit of time T times smaller and the unit of mass M times smaller, the numbers h, t_c and m grow by the factors L, T and M, the velocity v_c by LT^{-1} and the acceleration g by LT^{-2}. These factors are the **dimensions** of the quantities:

$$[h] = L, \qquad [t_c] = T, \qquad [m] = M, \qquad [v_c] = LT^{-1}, \qquad [g] = LT^{-2}$$

The dimensions are always power-law monomials $L^\alpha T^\beta M^\gamma$, with α, β, γ rational numbers (Notes.pdf, p. 58).

The two dimensionless ratios

Changing the units of length and time changes $(h/g)^{1/2}$ by a factor T and $(gh)^{1/2}$ by a factor LT^{-1}; check: $[(h/g)^{1/2}] = (L \cdot L^{-1}T^2)^{1/2} = T$ and $[(gh)^{1/2}] = (LT^{-2} \cdot L)^{1/2} = LT^{-1}$. Hence the ratios (Notes.pdf, Eqs. 6.2-6.3)

$$\Pi_1 = \frac{t_c}{(h/g)^{1/2}} \tag{6.2}$$

$$\Pi_2 = \frac{v_c}{(gh)^{1/2}} \tag{6.3}$$

have dimension $[\Pi_1] = [\Pi_2] = 1$: their value is the same in every system of units. With (6.1) they become (Notes.pdf, Eqs. 6.4-6.5)

$$\Pi_1 = \frac{t_c(m,h,g)}{(h/g)^{1/2}} = \Pi_1(m,h,g) \tag{6.4}$$

$$\Pi_2 = \frac{v_c(m,h,g)}{(gh)^{1/2}} = \Pi_2(m,h,g) \tag{6.5}$$

Removing the parameters one by one

The key argument of the whole chapter (Notes.pdf, p. 59):

1. Change **only the unit of mass**. The number m changes arbitrarily while h and g stay fixed, and Π_1, Π_2 (dimensionless) cannot change. So Π_1 and Π_2 do not depend on m: $\Pi_1 = \Pi_1(h,g)$, $\Pi_2 = \Pi_2(h,g)$.
2. Change **only the unit of time**. The number g is multiplied by T^{-2} while h stays fixed; again Π_1, Π_2 do not change, so they do not depend on g.
3. Change **the unit of length**. h changes arbitrarily and the Π's do not, so they do not depend on h either.

What is left is (Notes.pdf, Eqs. 6.6-6.7)

$$\frac{t_c}{(h/g)^{1/2}} = C_1 \tag{6.6}$$

$$\frac{v_c}{(gh)^{1/2}} = C_2 \tag{6.7}$$

Dimensional analysis gives the form of the law, not the constants. The integration above gives $C_1 = C_2 = \sqrt{2}$; a **single experiment** (drop one object, measure t_c and v_c) would give them too, and then (6.6)-(6.7) hold for any mass, any height and any planet.

2. Physical Quantities with Independent Dimensions

How many fundamental units

A geometric problem needs one unit (length), a kinematic problem two (length, time), mechanics three (mass, length, time); thermal problems add temperature Θ, electromagnetic ones electric charge. The chapter works with mechanical problems, $[a] = L^{\alpha} T^{\beta} M^{\gamma}$ (Notes.pdf, p. 60).

Definition

The quantities $a_1, \ldots, a_k$ have **independent dimensions** when the dimension of none of them can be written as a product of powers of the dimensions of the other $k-1$.

Example: ρ , U , μ

Take $[\rho] = ML^{-3}$, $[U] = LT^{-1}$, $[\mu] = ML^{-1}T^{-1}$. Assume the opposite: that two numbers x, y exist with $[\rho] = [U]^x [\mu]^y$ (Notes.pdf, Eq. 6.8). Written with the exponents of (L, M, T):

$$L^{-3} M^{1} T^{0} = \left(L^{1} M^{0} T^{-1}\right)^{x} \left(L^{-1} M^{1} T^{-1}\right)^{y} \tag{6.8}$$

Equating the exponents of L, M and T (Notes.pdf, Eq. 6.9):

$$-3 = x - y, \qquad 1 = y, \qquad 0 = -x - y \tag{6.9}$$

The second equation gives $y = 1$; the first then gives $x = -3 + 1 = -2$; the third requires $0 = -x - y = 2 - 1 = 1$, which is false. No pair (x, y) exists, so ρ, U, μ have independent dimensions. By contrast, p, ρ, U do not: $[\rho][U]^2 = ML^{-3} \cdot L^2 T^{-2} = ML^{-1}T^{-2} = [p]$, that is $x = 1, y = 2$.

The rank test

The equations (6.9) say that the exponent vector of ρ is a combination of those of U and μ (Notes.pdf, Eq. 6.10):

$$(-3, 1, 0) = x\,(1, 0, -1) + y\,(-1, 1, -1) \tag{6.10}$$

so $a_1, \ldots, a_k$ have independent dimensions if and only if **the rank of the matrix of exponents equals** k. Rows ρ, U, μ; columns L, M, T (Notes.pdf, Eq. 6.11):

$$\begin{pmatrix} -3 & 1 & 0 \\ 1 & 0 & -1 \\ -1 & 1 & -1 \end{pmatrix} \tag{6.11}$$

Expanding along the first row: $\det = -3\,[0 \cdot (-1) - (-1) \cdot 1] - 1\,[1 \cdot (-1) - (-1) \cdot (-1)] + 0 = -3(1) - 1(-2) = -1 \neq 0$, so the rank is 3. For ρ, U, p (Notes.pdf, Eq. 6.12):

$$\begin{pmatrix} -3 & 1 & 0 \\ 1 & 0 & -1 \\ -1 & 1 & -2 \end{pmatrix} \tag{6.12}$$

$\det = -3\,[0 \cdot (-2) - (-1) \cdot 1] - 1\,[1 \cdot (-2) - (-1) \cdot (-1)] = -3(1) - 1(-3) = 0$, and the first two rows are clearly not proportional, so the rank is 2: the three quantities are not independent.

> EXAM PRACTICE
>
> With three fundamental units the matrix has three columns, so **at most three** quantities can have independent dimensions. To find k, look for the largest set of rows with a non-zero determinant (a 3×3 minor first; if all vanish, a 2×2 one).

3. The Π Theorem (Vaschy-Buckingham Theorem)

The general problem

We look for an unknown quantity a_0 that depends on n governing parameters (Notes.pdf, Eq. 6.13):

$$a_0 = f(a_1, \ldots, a_k, a_{k+1}, \ldots, a_n) \tag{6.13}$$

Let $k \leq n$ be the largest number of parameters with independent dimensions, and $a_1, \ldots, a_k$ one such set. The dimensions of every other parameter are products of powers of theirs (Notes.pdf, Eq. 6.14):

$$[a_{k+1}] = [a_1]^{\alpha_{k+1}} \cdots [a_k]^{\gamma_{k+1}}, \quad [a_{k+2}] = [a_1]^{\alpha_{k+2}} \cdots [a_k]^{\gamma_{k+2}}, \quad \ldots, \quad [a_n] = [a_1]^{\alpha_n} \cdots [a_k]^{\gamma_n} \tag{6.14}$$

In general $0 < k < n$; $k = n$ happens in the falling body. The same holds for the unknown (Notes.pdf, Eq. 6.15):

$$[a_0] = [a_1]^{\alpha_0} \cdots [a_k]^{\gamma_0} \tag{6.15}$$

> **TWO TYPING SLIPS IN THE NOTES (P. 61)**
> In (6.14) the notes print the exponent of $[a_k]$ in the second line as γ_{k+1}; it must be γ_{k+2}, as written above. The sentence after (6.14) reads "or $k = n$, as occurs if all parameters $a_1, \ldots, a_n$ are dimensionless"; for dimensionless parameters no unit can change them, so the intended case is $k = 0$.

The dimensionless groups

Dividing each quantity by the product of powers with its own dimensions gives $n - k + 1$ dimensionless numbers (Notes.pdf, Eq. 6.16):

$$\Pi_0 = \frac{a_0}{a_1^{\alpha_0} \cdots a_k^{\gamma_0}}, \quad \Pi_1 = \frac{a_{k+1}}{a_1^{\alpha_{k+1}} \cdots a_k^{\gamma_{k+1}}}, \quad \Pi_2 = \frac{a_{k+2}}{a_1^{\alpha_{k+2}} \cdots a_k^{\gamma_{k+2}}}, \quad \ldots, \quad \Pi_{n-k} = \frac{a_n}{a_1^{\alpha_n} \cdots a_k^{\gamma_n}} \tag{6.16}$$

Dividing (6.13) by $a_1^{\alpha_0} \cdots a_k^{\gamma_0}$ and writing $a_{k+1}, \ldots, a_n$ through the Π's, $a_{k+j} = \Pi_j \, a_1^{\alpha_{k+j}} \cdots a_k^{\gamma_{k+j}}$ (Notes.pdf, Eq. 6.17):

$$\Pi_0 = \frac{f(a_1, \ldots, a_k, a_{k+1}, \ldots, a_n)}{a_1^{\alpha_0} \cdots a_k^{\gamma_0}} = \frac{f\left(a_1, \ldots, a_k, \Pi_1 a_1^{\alpha_{k+1}} \cdots a_k^{\gamma_{k+1}}, \ldots, \Pi_{n-k} a_1^{\alpha_n} \cdots a_k^{\gamma_n}\right)}{a_1^{\alpha_0} \cdots a_k^{\gamma_0}} \tag{6.17}$$

that is, Π_0 is a function of $a_1, \ldots, a_k$ and of the Π's (Notes.pdf, Eq. 6.18):

$$\Pi_0 = \Pi_0(a_1, \ldots, a_k, \Pi_1, \ldots, \Pi_{n-k}) \tag{6.18}$$

Worked example: pressure loss in a pipe

The pressure loss per unit length of a liquid flowing along a long circular pipe depends on the volume flux Q, the diameter D, the density ρ and the viscosity μ (Notes.pdf, Eq. 6.19):

$$P_l = -\frac{\partial p}{\partial x} = f(Q, D, \rho, \mu) \tag{6.19}$$

Here $n = 4$. Dimensions: $[Q] = L^3T^{-1}$, $[D] = L$, $[\rho] = ML^{-3}$, $[\mu] = ML^{-1}T^{-1}$, $[P_l] = [p]/L = ML^{-2}T^{-2}$. The set (Q, D, ρ) is independent: ρ is the only one with M, Q the only one with T, and D adds L; so $k = 3$.

Exponents of μ. Write $[\mu] = [Q]^a[D]^b[\rho]^c$ and equate exponents:

$$M:\ 1 = c, \qquad T:\ -1 = -a, \qquad L:\ -1 = 3a + b - 3c$$

so $c = 1$, $a = 1$ and $b = -1 - 3(1) + 3(1) = -1$ (Notes.pdf, Eq. 6.20):

$$[\mu] = [Q]^1[D]^{-1}[\rho]^1 \tag{6.20}$$

Exponents of P_l. With $[P_l] = [Q]^a[D]^b[\rho]^c$:

$$M:\ 1 = c, \qquad T:\ -2 = -a, \qquad L:\ -2 = 3a + b - 3c$$

so $c = 1$, $a = 2$ and $b = -2 - 6 + 3 = -5$ (Notes.pdf, Eq. 6.21):

$$[P_l] = [Q]^2[D]^{-5}[\rho]^1 \tag{6.21}$$

The two dimensionless groups are (Notes.pdf, Eq. 6.22)

$$\Pi_0 = \frac{P_l}{\rho Q^2/D^5} \qquad \text{and} \qquad \Pi_1 = \frac{\mu}{\rho Q/D} \tag{6.22}$$

and substituting $\mu = \Pi_1\, \rho Q/D$ into (6.19) gives the form (6.18) (Notes.pdf, Eq. 6.23):

$$\Pi_0 = \frac{f(Q, D, \rho, \Pi_1 \rho Q/D)}{\rho Q^2/D^5} = F(Q, D, \rho, \Pi_1) \tag{6.23}$$

Now the argument of Section 1, repeated:

1. Make the unit of mass M times smaller: ρ grows by M, while Π_0, Π_1, Q, D do not change. So F does not depend on ρ: $\Pi_0 = F(Q, D, \Pi_1)$.
2. Make the unit of time T times smaller: the number Q changes by the factor T^{-1} (since $[Q] = L^3T^{-1}$), while D, Π_0, Π_1 do not change: $\Pi_0 = F(D, \Pi_1)$.
3. Change the unit of length: $\Pi_0 = F(\Pi_1)$.

The result (Notes.pdf, Eq. 6.24):

$$\frac{P_l}{\rho Q^2/D^5} = F\left(\frac{\mu}{\rho Q/D}\right) \tag{6.24}$$

The theorem

Repeating the argument on (6.18): a change of units can always modify a_1 alone, leaving $a_2, \ldots, a_k$ and all the Π's fixed, so Π_0 does not depend on a_1; one by one, it does not depend on $a_2, \ldots, a_k$ either (Notes.pdf, Eq. 6.25):

$$\Pi_0 = F(\Pi_1, \ldots, \Pi_{n-k}) \tag{6.25}$$

> **THE Π THEOREM (NOTES.PDF, P. 62)**
> If an unknown a_0 depends on n governing parameters $a_1, \ldots, a_n$, the number of parameters can always be reduced to $n - k$, where k is the number of governing parameters with independent dimensions. The k quantities with independent dimensions are used to write the other $n - k$ parameters and the unknown in dimensionless form, which gives (6.25).

Why it matters: fewer experiments

For the pipe, the theorem reduces four parameters to one. A single series of experiments in one pipe with one fluid, varying Q and measuring P_l, plotted as Π_0 against Π_1, gives the law for **any fluid in any pipe**. For a new case (Q^*, ρ^*, μ^*, D^*): compute $\Pi_1^* = \mu^*/(\rho^* Q^*/D^*)$, read Π_0^* on the curve, and solve $P_l^* = \Pi_0^*\,\rho^*(Q^*)^2/(D^*)^5$ (Notes.pdf, p. 63).

4. Taylor's Analysis of a Nuclear Explosion

Governing parameters

An atomic explosion releases a large energy E almost instantaneously at one point; a strong spherical shock of radius r_o grows with time t. In the first stages the pressure p_o behind the shock is thousands of times the ambient pressure, so the ambient pressure does not enter, and (6.13) becomes (Notes.pdf, Eq. 6.26):

$$r_o = f_1(E, \rho_a, t) \qquad \text{and} \qquad p_o = f_2(E, \rho_a, t) \tag{6.26}$$

Rank

Rows E, ρ_a, t; columns L, M, T, from $[E] = ML^2T^{-2}$, $[\rho_a] = ML^{-3}$, $[t] = T$ (Notes.pdf, Eq. 6.27):

$$\begin{pmatrix} 2 & 1 & -2 \\ -3 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \tag{6.27}$$

Expanding along the third row: $\det = 1\cdot(2\cdot 1 - 1\cdot(-3)) = 5 \neq 0$. So $k = 3 = n$ and **no governing parameter is left**: $n - k = 0$.

The exponents

Radius. $[r_o] = L = [E]^a[\rho_a]^b[t]^c$:

$$M:\ 0 = a + b, \qquad L:\ 1 = 2a - 3b, \qquad T:\ 0 = -2a + c$$

From the first, $b = -a$; the second becomes $1 = 2a + 3a = 5a$, so $a = 1/5$, $b = -1/5$; the third gives $c = 2a = 2/5$.

Pressure. $[p_o] = ML^{-1}T^{-2} = [E]^a[\rho_a]^b[t]^c$:

$$M:\ 1 = a + b, \qquad L:\ -1 = 2a - 3b, \qquad T:\ -2 = -2a + c$$

With $b = 1 - a$: $-1 = 2a - 3 + 3a$, so $a = 2/5$, $b = 3/5$, and $c = -2 + 2a = -6/5$.

The solution is therefore (Notes.pdf, Eqs. 6.28-6.29)

$$\frac{r_o}{E^{1/5}\rho_a^{-1/5}t^{2/5}} = C_1 \qquad \text{and} \qquad \frac{p_o}{E^{2/5}\rho_a^{3/5}t^{-6/5}} = C_2 \tag{6.28}$$

$$r_o = C_1 E^{1/5}\rho_a^{-1/5}t^{2/5} \qquad \text{and} \qquad p_o = C_2 E^{2/5}\rho_a^{3/5}t^{-6/5} \tag{6.29}$$

C_1 and C_2 come from integrating the compressible-flow equations, or from measurements.

The shock slows down, and when the law stops holding

The shock speed is the time derivative of (6.29), with the power rule $\frac{d}{dt}t^{2/5} = \frac{2}{5}t^{2/5-1} = \frac{2}{5}t^{-3/5}$:

$$\frac{dr_o}{dt} = \frac{2}{5}\,C_1 E^{1/5}\rho_a^{-1/5}t^{-3/5}$$

Both the speed and p_o decrease with time. The law fails when p_o falls to the order of the ambient pressure p_a. Setting $E^{2/5}\rho_a^{3/5}t^{-6/5} \sim p_a$ and solving for t: $t^{6/5} \sim E^{2/5}\rho_a^{3/5}/p_a$, and raising to $5/6$,

$$t \sim \frac{E^{1/3}\rho_a^{1/2}}{p_a^{5/6}}$$

(Notes.pdf, p. 64). The same reasoning applies to the cylindrical shock of a lightning stroke, with E the energy per unit length, $[E] = MLT^{-2}$.

5. Parametric Dependence of the Solution

The hard step is the list of parameters

Missing an important parameter, or adding spurious ones, gives a wrong result. In fluid mechanics the list need not be a guess: **write the conservation equations with their initial and boundary conditions**, and read the parameters from them (Notes.pdf, p. 64).

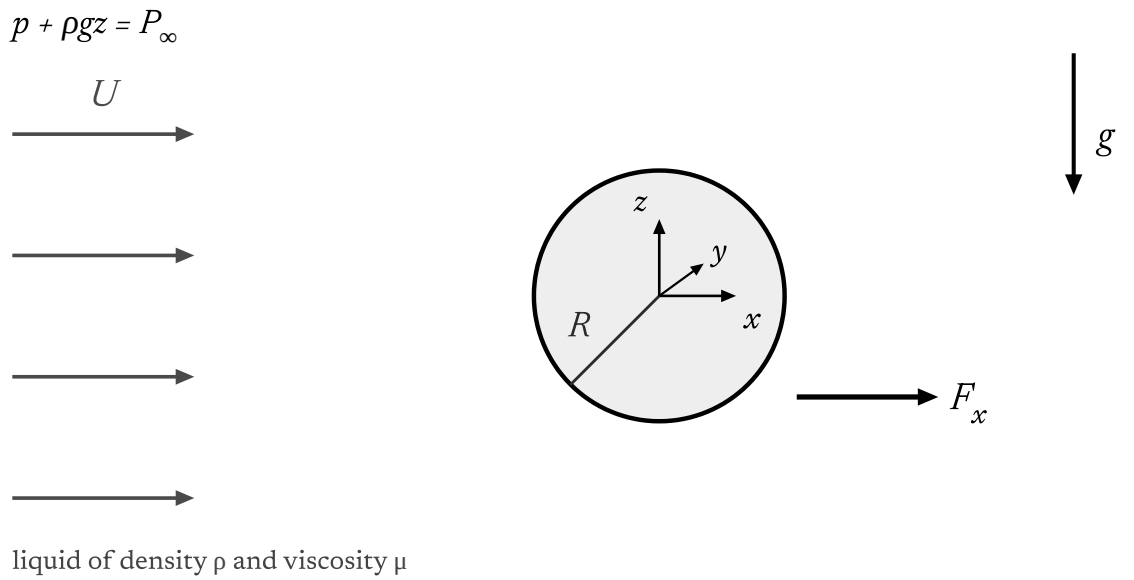


Figure 2. Fluid motion around a sphere, Fig. 6.2 of the notes, seen from the sphere: far away the liquid moves with velocity U and its pressure is hydrostatic, p + ρgz = P_∞; the unknown is the drag F_x.

A first guess for the sphere

A sphere of radius R moves with constant velocity U in a stagnant liquid of density ρ and viscosity μ; far away the pressure is hydrostatic, $p + \rho g z = P_\infty$. A first guess for the drag (Notes.pdf, Eq. 6.30):

$$F_x = F_x(R, U, \rho, \mu, P_\infty, g) \tag{6.30}$$

Exponent matrix, rows $R, U, \rho, \mu, P_\infty, g$; columns L, M, T (Notes.pdf, Eq. 6.31):

$$\begin{pmatrix} 1 & 0 & 0 \\ 1 & 0 & -1 \\ -3 & 1 & 0 \\ -1 & 1 & -1 \\ -1 & 1 & -2 \\ 1 & 0 & -2 \end{pmatrix} \tag{6.31}$$

The rank of a 6×3 matrix is at most 3, and the first three rows give $\det = 1 \cdot (0 \cdot 0 - (-1) \cdot 1) - 0 + 0 = 1 \neq 0$, so $k = 3$ with (R, U, ρ). Since $[F_x] = MLT^{-2} = [\rho][U]^2[R]^2$, the dimensionless force is (Notes.pdf, Eq. 6.32)

$$\Pi_0 = \frac{F_x}{\rho U^2 R^2} \tag{6.32}$$

Check the other three: $[\rho][U][R] = ML^{-3} \cdot LT^{-1} \cdot L = ML^{-1}T^{-1} = [\mu]$; $[\rho][U]^2 = ML^{-1}T^{-2} = [P_\infty]$; $[U]^2[R]^{-1} = LT^{-2} = [g]$. So (Notes.pdf, Eq. 6.33)

$$\Pi_1 = \frac{\mu}{\rho U R}, \qquad \Pi_2 = \frac{P_\infty}{\rho U^2}, \qquad \Pi_3 = \frac{g}{U^2/R} \tag{6.33}$$

and the theorem gives (Notes.pdf, Eq. 6.34)

$$\frac{F_x}{\rho U^2 R^2} = f\left(\frac{\mu}{\rho U R}, \frac{P_\infty}{\rho U^2}, \frac{g}{U^2/R}\right) \tag{6.34}$$

Six parameters became three, but **(6.34) is wrong**: P_∞ and g do not influence the drag. With the correct list $F_x = F_x(R, U, \rho, \mu)$ the result is (Notes.pdf, Eq. 6.35)

$$\frac{F_x}{\rho U^2 R^2} = f\left(\frac{\mu}{\rho U R}\right) \tag{6.35}$$

Reading the parameters from the equations

In the frame of the sphere the liquid satisfies (Notes.pdf, Eq. 6.36)

$$\nabla \cdot \bar{v} = 0, \quad \rho\, \bar{v} \cdot \nabla \bar{v} = -\nabla p + \mu \nabla^2 \bar{v} - \rho g\, \bar{e}_z; \qquad |\bar{x}| \to \infty : \; p = P_\infty - \rho g z, \; \bar{v} = U \bar{e}_x; \qquad |\bar{x}| = R : \; \bar{v} = 0 \tag{6.36}$$

and the force on the sphere is the integral of the pressure and of the viscous stress $\bar{\bar{\tau}}' = \mu(\nabla \bar{v} + \nabla \bar{v}^T)$ over its surface (Notes.pdf, Eqs. 6.37-6.38):

$$\bar{F} = -\int_{|\bar{x}|=R} p\,\bar{n}\,d\sigma + \int_{|\bar{x}|=R} \mu(\nabla\bar{v} + \nabla\bar{v}^T)\cdot\bar{n}\,d\sigma \tag{6.37}$$

$$F_x = \bar{F}\cdot\bar{e}_x = -\int_{|\bar{x}|=R} p\,n_x\,d\sigma + \int_{|\bar{x}|=R} \mu\left[(\nabla\bar{v} + \nabla\bar{v}^T)\cdot\bar{n}\right]\cdot\bar{e}_x\,d\sigma \tag{6.38}$$

At first sight P_∞ and g do appear. Two observations remove them:

1. Gravity is a gradient, $-\nabla p - \rho g\bar{e}_z = -\nabla(p + \rho g z)$, so it can be absorbed in the **reduced pressure**.
2. Only pressure differences move the fluid; the level P_∞ does not matter.

Both suggest $P' = p + \rho g z - P_\infty$. In terms of P' (Notes.pdf, Eq. 6.39):

$$\nabla\cdot\bar{v} = 0, \quad \rho\,\bar{v}\cdot\nabla\bar{v} = -\nabla P' + \mu\nabla^2\bar{v}; \qquad |\bar{x}|\to\infty:\ P' = 0,\ \bar{v} = U\bar{e}_x; \qquad |\bar{x}| = R:\ \bar{v} = 0 \tag{6.39}$$

Neither P_∞ nor g appears, so $P'(\bar{x})$ and $\bar{v}(\bar{x})$ do not depend on them. Substituting $p = P' + P_\infty - \rho g z$ in (6.37), and using $\int P_\infty \bar{n}\,d\sigma = 0$ (a constant pressure on a closed surface gives no force) (Notes.pdf, Eq. 6.40):

$$\bar{F} = -\int_{|\bar{x}|=R} P'\,\bar{n}\,d\sigma + \int_{|\bar{x}|=R} \mu(\nabla\bar{v} + \nabla\bar{v}^T)\cdot\bar{n}\,d\sigma + \int_{|\bar{x}|=R} \rho g z\,\bar{n}\,d\sigma \tag{6.40}$$

The last term is Archimedes' buoyancy. By the Gauss theorem over the volume V of the sphere ($\bar{n}$ pointing out of the sphere) and $\nabla(\rho g z) = \rho g\,\bar{e}_z$ (Notes.pdf, Eq. 6.41):

$$\int_{|\bar{x}|=R} \rho g z\,\bar{n}\,d\sigma = \int_V \nabla(\rho g z)\,dV = \rho g\,\frac{4}{3}\pi R^3\,\bar{e}_z \tag{6.41}$$

It is vertical, so it does not contribute to the drag (Notes.pdf, Eq. 6.42):

$$F_x = \bar{F}\cdot\bar{e}_x = -\int_{|\bar{x}|=R} P'\,n_x\,d\sigma + \int_{|\bar{x}|=R} \mu\left[(\nabla\bar{v} + \nabla\bar{v}^T)\cdot\bar{n}\right]\cdot\bar{e}_x\,d\sigma \tag{6.42}$$

This proves that $F_x = F_x(R, U, \rho, \mu)$ and that (6.35), not (6.34), is the right result.

When pressure level and gravity do matter

- **Gases**: the absolute pressure enters through the equation of state $p = \rho R_g T$.
- **Liquids with a free surface that is not flat** (waves): the condition $p = p_a$ on the unknown surface $z_s(x, y)$ brings g back. A sphere moving at a depth of order R below the surface feels a drag that depends on g and on the depth, and the reduced pressure no longer removes g (Notes.pdf, p. 66). This is why g appears in the ship, tsunami and wave-maker problems (DA1, DA6, DA11).

The repeating set is not unique

Choosing (U, R, μ) instead of (U, R, ρ) gives (Notes.pdf, Eq. 6.43)

$$\frac{F_x}{\mu U R} = f\left(\frac{\rho U R}{\mu}\right) \tag{6.43}$$

Both (6.35) and (6.43) are valid. In both, the dimensionless force depends on the **Reynolds number** (Notes.pdf, Eq. 6.44):

$$\mathrm{Re} = \frac{\rho U R}{\mu} \tag{6.44}$$

6. Nondimensionalization of the Navier-Stokes Equations

Scales from the boundary conditions

Instead of the Π theorem, the problem itself can be written in dimensionless variables. For the sphere, the boundary conditions give the velocity scale U and the length scale R: $\bar{v}^* = \bar{v}/U$, $\bar{x}^* = \bar{x}/R$, hence $\nabla^* = R\nabla$ and $d\sigma^* = R^{-2} d\sigma$ (Notes.pdf, p. 67). The scale Δp of P' is not obvious: it depends on the kind of motion.

Order of magnitude of each term

Velocities are of order U and change by $O(U)$ over distances $O(R)$, so $O(\nabla \bar{v}) \sim U/R$ and $O(\nabla^2 \bar{v}) \sim U/R^2$ (Notes.pdf, Eq. 6.45):

$$\underbrace{\rho\, \bar{v} \cdot \nabla \bar{v}}_{\rho U^2/R} = \underbrace{-\nabla P'}_{\Delta p/R} + \underbrace{\mu \nabla^2 \bar{v}}_{\mu U/R^2} \tag{6.45}$$

The pressure gradient is almost always essential: without it, (6.39) would have three unknowns (the velocity components) and four equations (continuity and three momentum components). The ratio of the other two terms is (Notes.pdf, Eq. 6.46)

$$\frac{O(\rho\,\bar v\cdot\nabla\bar v)}{O(\mu\nabla^2\bar v)}=\frac{\rho U^2/R}{\mu U/R^2}=\frac{\rho U R}{\mu}\tag{6.46}$$

the Reynolds number (6.44). Three regimes (Notes.pdf, p. 67):

REGIME	BALANCE IN (6.45)	PRESSURE SCALE
$\mathrm{Re}\sim 1$	all three terms	either estimate
$\mathrm{Re}\gg 1$	$\rho\,\bar v\cdot\nabla\bar v=-\nabla P'$ (viscosity negligible)	$\Delta p\sim\rho U^2$
$\mathrm{Re}\ll 1$	$0=-\nabla P'+\mu\nabla^2\bar v$ (acceleration negligible)	$\Delta p\sim\mu U/R$

Scaling for $\mathrm{Re}\gtrsim 1$

With $p^*=(p+\rho g z-P_\infty)/(\rho U^2)$, substitute $\bar v=U\bar v^*$, $\bar x=R\bar x^*$, $\nabla=\nabla^*/R$, $P'=\rho U^2 p^*$ in each term of (6.39):

$$\rho\,\bar v\cdot\nabla\bar v=\frac{\rho U^2}{R}\,\bar v^*\cdot\nabla^*\bar v^*,\qquad \nabla P'=\frac{\rho U^2}{R}\nabla^* p^*,\qquad \mu\nabla^2\bar v=\frac{\mu U}{R^2}\nabla^{*2}\bar v^*$$

Dividing by $\rho U^2/R$, the viscous term carries $\mu U/R^2\div\rho U^2/R=\mu/(\rho U R)=1/\mathrm{Re}$. The problem (6.39) becomes (Notes.pdf, Eq. 6.47)

$$\nabla^*\cdot\bar v^*=0,\quad \bar v^*\cdot\nabla^*\bar v^*=-\nabla^* p^*+\frac{1}{\mathrm{Re}}\nabla^{*2}\bar v^*;\qquad |\bar x^*|\to\infty:\ p^*=0,\ \bar v^*=\bar e_x;\qquad |\bar x^*|=1:\ \bar v^*=0\tag{6.47}$$

In the force (6.42), $P'\,d\sigma=\rho U^2R^2\,p^*\,d\sigma^*$ and the viscous stress is $\mu(U/R)(\nabla^*\bar v^*+\nabla^*\bar v^{*T})$, which times $d\sigma=R^2 d\sigma^*$ gives $\mu U R\,(\dots)\,d\sigma^*$. Dividing by ρU^2R^2 (Notes.pdf, Eq. 6.48):

$$\frac{F_x}{\rho U^2R^2}=-\int_{|\bar x^*|=1}p^*\,\bar n\cdot\bar e_x\,d\sigma^*+\frac{1}{\mathrm{Re}}\int_{|\bar x^*|=1}\left[(\nabla^*\bar v^*+\nabla^*\bar v^{*T})\cdot\bar n\right]\cdot\bar e_x\,d\sigma^*\tag{6.48}$$

The only parameter left is Re: $F_x/(\rho U^2R^2)$ is a function of Re alone, exactly (6.35).

REFERENCE IN THE NOTES (P. 68)

The notes say that this scaling reduces "(6.40)"; the problem being scaled is the system (6.39) (the field equations and boundary conditions), while the force is (6.42). The same applies to the sentence before (6.49).

Scaling for $\mathrm{Re} \lesssim 1$

With $p^* = (p + \rho g z - P_\infty)/(\mu U/R)$, now $\nabla P' = (\mu U/R^2)\nabla^* p^*$; dividing every term by $\mu U/R^2$, the convective term carries $\rho U^2/R \div \mu U/R^2 = \mathrm{Re}$ (Notes.pdf, Eq. 6.49):

$$\nabla^* \cdot \bar{v}^* = 0, \quad \mathrm{Re}\, \bar{v}^* \cdot \nabla^* \bar{v}^* = -\nabla^* p^* + \nabla^{*2} \bar{v}^*; \qquad |\bar{x}^*| \to \infty : \; p^* = 0, \; \bar{v}^* = \bar{e}_x; \qquad |\bar{x}^*| = 1 : \; \bar{v}^* = 0 \tag{6.49}$$

and the force, divided by $\mu U R$ (Notes.pdf, Eq. 6.50):

$$\frac{F_x}{\mu U R} = -\int_{|\bar{x}^*|=1} p^* \, \bar{n} \cdot \bar{e}_x \, d\sigma^* + \int_{|\bar{x}^*|=1} \left[(\nabla^* \bar{v}^* + \nabla^* \bar{v}^{*T}) \cdot \bar{n} \right] \cdot \bar{e}_x \, d\sigma^* \tag{6.50}$$

so $F_x/(\mu U R)$ is a function of Re only, the result (6.43).

Π THEOREM OR NONDIMENSIONAL EQUATIONS?

Both give the same groups, but the equations tell you **which parameters really enter** (no spurious P_∞ or g) and **which terms are small** in each limit. The Π theorem alone cannot do either. Problem DA14 compares the two methods on the rotating cavity.

7. Selection of Parameters with Independent Dimensions

The rule

The set $a_1, \ldots, a_k$ should contain parameters with a **significant influence** for the flow considered (Notes.pdf, p. 68). For the sphere, R and U obviously matter. The third one:

- if viscosity is important, μ, which leads to (6.43);
- if viscosity is not expected to matter, ρ, which leads to (6.35).

Evaluate $\mathrm{Re} = \rho U_c L_c/\mu$ with the characteristic velocity and size: $\mathrm{Re} \lesssim 1$ suggests (U_c, L_c, μ); $\mathrm{Re} \gtrsim 1$ suggests (U_c, L_c, ρ).

The two extreme limits

For $\mathrm{Re} \gg 1$ the $1/\mathrm{Re}$ term of (6.47)-(6.48) disappears, so the dimensionless force no longer depends on Re (Notes.pdf, Eq. 6.51):

$$\lim_{\mathrm{Re}\gg 1} \frac{F_x}{\rho U^2 R^2} = C_1 \tag{6.51}$$

For $\mathrm{Re} \ll 1$ the Re term of (6.49) disappears (Notes.pdf, Eq. 6.52):

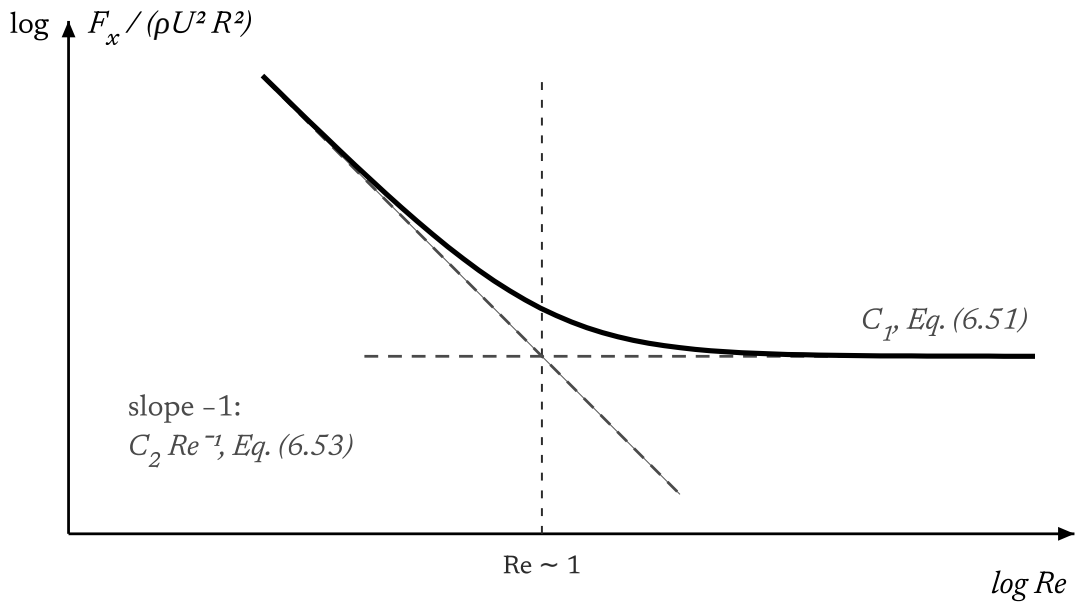
$$\lim_{\mathrm{Re}\ll 1} \frac{F_x}{\mu U R} = C_2 \tag{6.52}$$

With the "wrong" set, the limit still depends on Re. Since $\frac{F_x}{\rho U^2 R^2} = \frac{F_x}{\mu U R} \cdot \frac{\mu U R}{\rho U^2 R^2} = \frac{F_x}{\mu U R} \cdot \frac{1}{\mathrm{Re}}$, (6.52) gives (Notes.pdf, Eq. 6.53)

$$\lim_{\mathrm{Re}\ll 1} \frac{F_x}{\rho U^2 R^2} = C_2\,\mathrm{Re}^{-1} \tag{6.53}$$

and, since $\frac{F_x}{\mu U R} = \frac{F_x}{\rho U^2 R^2} \cdot \mathrm{Re}$, (6.51) gives (Notes.pdf, Eq. 6.54)

$$\lim_{\mathrm{Re}\gg 1} \frac{F_x}{\mu U R} = C_1\,\mathrm{Re} \tag{6.54}$$



Sketch of the two limits only: the constants C_1 and C_2 are not given in the notes

Figure 3. Sketch, on logarithmic axes, of the two limits of Eq. (6.35): for Re ≪ 1 the dimensionless drag falls as C_2/Re, Eq. (6.53); for Re ≫ 1 it tends to the constant C_1, Eq. (6.51). Only the slopes are drawn: the constants and the shape of the curve in between are not in the notes.

In dimensional terms: for $\mathrm{Re} \ll 1$, $F_x = C_2\, \mu U R$ (drag linear in U, independent of ρ); for $\mathrm{Re} \gg 1$, $F_x = C_1\, \rho U^2 R^2$ (drag quadratic in U, independent of μ).

> **TYPING SLIP IN THE NOTES (P. 69)**
> Before (6.51) the notes write "$F_x/(\rho U R^2)$ becomes independent of Re"; the quantity is $F_x/(\rho U^2 R^2)$, as in (6.51).

General form of the limits

When one group of (6.25) takes extreme values, say $\Pi_1 \gg 1$, two things can happen (Notes.pdf, Eqs. 6.55-6.56):

$$\lim_{\Pi_1 \gg 1} \Pi_0(\Pi_1, \ldots, \Pi_{n-k}) = f(\Pi_2, \ldots, \Pi_{n-k}) \tag{6.55}$$

$$\lim_{\Pi_1 \gg 1} \Pi_0(\Pi_1, \ldots, \Pi_{n-k}) = \Pi_1^{\lambda}\, f(\Pi_2, \ldots, \Pi_{n-k}) \tag{6.56}$$

In (6.55) one parameter disappears; in (6.56) it survives only as a power, with λ to be found. A good choice of $a_1, \dots, a_k$ usually gives the simpler form (6.55).

> EXAM PRACTICE
> In the problems, "simplify assuming the viscosity has a negligible effect" means: use ρ (not μ) in the repeating set, then drop the group that contains μ. Always finish by **checking** the assumption with numbers: $\mathrm{Re} \gg 1$ for both the model and the real system (DA1, DA6, DA12).

8. Physical Similarity

Designing an experiment

A full-scale phenomenon often cannot be reproduced in the laboratory. Example of the notes: the drag F_1 on a sphere of radius R_1 moving at U_1 in an oil (ρ_1, μ_1), to be measured in a water tunnel (ρ_2, μ_2) with the only available spheres, of radius $R_2 \neq R_1$ (Notes.pdf, p. 70).

By (6.35), $F_x/(\rho U^2 R^2)$ depends only on Re. If the experiment has the same Reynolds number, $\rho_2 U_2 R_2/\mu_2 = \rho_1 U_1 R_1/\mu_1$, solving for U_2 (Notes.pdf, Eq. 6.57):

$$U_2 = \frac{\mu_2}{\mu_1}\,\frac{\rho_1}{\rho_2}\,\frac{R_1}{R_2}\,U_1 \tag{6.57}$$

Then the dimensionless forces are equal (Notes.pdf, Eq. 6.58):

$$\frac{F_1}{\rho_1 U_1^2 R_1^2} = \frac{F_2}{\rho_2 U_2^2 R_2^2} \tag{6.58}$$

and the measured F_2 gives the real force (Notes.pdf, Eq. 6.59):

$$F_1 = \frac{\rho_1 U_1^2 R_1^2}{\rho_2 U_2^2 R_2^2}\,F_2 \tag{6.59}$$

The general rule

For $\Pi_0 = F(\Pi_1, \dots, \Pi_{n-k})$, the model does not have to reproduce $a_1, \dots, a_n$. It is **physically similar** to the real system when every governing group $\Pi_1, \dots, \Pi_{n-k}$ takes the same value in both; then Π_0 is equal too, and the measured a_0 is converted to the real one by equating Π_0 (Notes.pdf, p. 71).

When not all groups can be matched, similarity is **partial**: match the groups that dominate the problem and accept different values of the secondary ones (Notes.pdf, p. 71).

Extension: why Reynolds and Froude cannot both be matched with the same fluid

This subsection is an **extension** used in problems DA1 and DA3. A ship or a stirred free surface depends on $\mathrm{Re} = UL/\nu$ (with $\nu = \mu/\rho$) and on the Froude number $\mathrm{Fr} = U/\sqrt{gL}$. Model m, prototype p, scale $\lambda = L_m/L_p$, same g:

$$\mathrm{Fr}_m = \mathrm{Fr}_p \implies \frac{U_m}{U_p} = \sqrt{\frac{L_m}{L_p}} = \lambda^{1/2}, \qquad \mathrm{Re}_m = \mathrm{Re}_p \implies \frac{U_m}{U_p} = \frac{\nu_m}{\nu_p}\frac{L_p}{L_m} = \frac{\nu_m}{\nu_p}\lambda^{-1}$$

Equating both velocity ratios: $\nu_m/\nu_p = \lambda^{3/2}$. With the same fluid ($\nu_m = \nu_p$) this forces $\lambda = 1$: no scaled model can match both. The two ways out are those of the problems:

- **neglect viscosity** when $\mathrm{Re} \gg 1$ in model and prototype, and match only Froude (DA1, DA6, DA11);
- **change the fluid** so that $\nu_m = \lambda^{3/2}\nu_p$ (DA3, tar replaced by water).

9. Practical Formulary of Topic 6

Dimensions in $[M, L, T, \Theta]$

QUANTITY	SYMBOL	DIMENSIONS
Velocity	U	LT^{-1}
Acceleration, gravity	g	LT^{-2}
Angular velocity, frequency	Ω, f	T^{-1}
Volume flux (3-D); per unit width (2-D)	Q; q	L^3T^{-1}; L^2T^{-1}
Density	ρ	ML^{-3}
Dynamic viscosity	μ	$ML^{-1}T^{-1}$
Kinematic viscosity	$\nu = \mu/\rho$	L^2T^{-1}
Pressure, stress	p, τ	$ML^{-1}T^{-2}$
Pressure gradient	$P_l = -\partial p/\partial x$	$ML^{-2}T^{-2}$
Force; force per unit length	F; F/L	MLT^{-2}; MT^{-2}
Energy, torque	E, M	ML^2T^{-2}
Power	W	ML^2T^{-3}
Speed of sound	c	LT^{-1}
Temperature (extension)	T	Θ
Thermal conductivity (extension)	k	$MLT^{-3}\Theta^{-1}$
Specific heat (extension)	c_p	$L^2T^{-2}\Theta^{-1}$

Dimensionless numbers

Only the Reynolds number is defined in Chapter 6; the others are **extensions**, given here because the problems use them.

NUMBER	DEFINITION	MEASURES	WHERE
Reynolds	$\mathrm{Re} = \rho UL/\mu = UL/\nu$	convection / viscous forces	Eq. (6.44), (6.46)
Froude (extension)	$\mathrm{Fr} = U/\sqrt{gL}$	inertia / gravity	DA1, DA3, DA6, DA11, DA13
Mach (extension)	$\mathrm{Ma} = U/c$	flow speed / speed of sound	DA9, DA12
Strouhal (extension)	$\mathrm{St} = fL/U$	unsteadiness / convection	DA2 (shedding period)
Euler (extension)	$\mathrm{Eu} = \Delta p/(\rho U^2)$	pressure differences / dynamic pressure	scaling of (6.47)
Prandtl (extension)	$\mathrm{Pr} = \mu c_p/k$	momentum / heat diffusion	energy equation
Eckert (extension)	$\mathrm{Ec} = U^2/(c_p\,\Delta T)$	kinetic energy / enthalpy differences	energy equation

Key results

SITUATION	RESULT	SOURCE
Independent dimensions	rank of the exponent matrix $= k$	Eqs. (6.10)-(6.12)
Π theorem	$a_0 = f(a_1, \ldots, a_n) \;\Rightarrow\; \Pi_0 = F(\Pi_1, \ldots, \Pi_{n-k})$	Eqs. (6.13)-(6.25)
$k = n$ (no group left)	$\Pi_0 = C$, constant	Eqs. (6.6)-(6.7), (6.28)
Pipe flow	$P_l D^5/(\rho Q^2) = F\left(\mu D/(\rho Q)\right)$	Eq. (6.24)
Strong explosion	$r_o = C_1 E^{1/5} \rho_a^{-1/5} t^{2/5}$, $p_o = C_2 E^{2/5} \rho_a^{3/5} t^{-6/5}$	Eq. (6.29)
Reduced pressure removes g and P_∞	$P' = p + \rho g z - P_\infty$ (not with a free surface)	Eq. (6.39)
Sphere drag	$F_x/(\rho U^2 R^2) = f(\mathrm{Re})$, or $F_x/(\mu U R) = f(\mathrm{Re})$	Eqs. (6.35), (6.43)
$\mathrm{Re} \gg 1$	$F_x = C_1 \rho U^2 R^2$; pressure scale ρU^2	Eqs. (6.47), (6.51)
$\mathrm{Re} \ll 1$	$F_x = C_2 \mu U R$; pressure scale $\mu U/R$	Eqs. (6.49), (6.52)
Similarity	equal $\Pi_1, \ldots, \Pi_{n-k} \Rightarrow$ equal Π_0	Eqs. (6.57)-(6.59)
Re and Fr with the same g (extension)	needs $\nu_m/\nu_p = (L_m/L_p)^{3/2}$	Section 8

Exam recipe

1. **List the parameters.** Write the unknown a_0 and everything it depends on. In doubt, write the equations and boundary conditions: geometry, velocities or rotation rates imposed at the boundaries, fluid properties that appear (ρ, μ; c for a gas at high speed; g only with a free surface or a weight). Count n.
2. **Build the exponent matrix** (rows: parameters; columns: M, L, T) and find k with a non-zero determinant.
3. **Choose the repeating set** of k parameters: those that clearly matter. Use ρ if viscosity is expected to be negligible ($\mathrm{Re} \gg 1$), μ if inertia is ($\mathrm{Re} \ll 1$). Never put the unknown in the set.
4. **Form each Π** by solving the system of exponents, one equation per fundamental unit, and check that each group is dimensionless.

5. **Simplify with the limits** stated in the problem: drop the group that contains the negligible parameter (6.55), or keep it as a power (6.56) when the set was not the natural one.
6. **Similarity.** Equate every remaining governing group between model and prototype to get the test conditions (velocity, rotation rate, fluid, mass). Equate Π_0 to convert the measurement. If two groups cannot both be matched, match the dominant one and check numerically that the neglected group is extreme in both systems.
7. **Tables of measurements.** Each row of a model table maps to a prototype row through the same scale factors (one for the independent variable, one for the unknown).
8. **Check** the dimensions of the final formula and its limits.